Framework for AI Adoption in Healthcare Sector: Integrated DELPHI, ISM–MICMAC Approach

Amit Kumar Gupta and Manoj Kumar Srivastava

***Abstract*—Artificial Intelligence (AI) adoption is transforming many industries, but its impact on the healthcare sector is life-changing. Recent studies and tests show that AI can deliver identical or better prognoses, diagnoses, and surgical outcomes than medical professionals. Healthcare sectors are adopting AI, and its applications are reforming it by reducing expenditure and exceeding patient satisfaction. The dearth of AI advocacy and adoption has forfeited large annual opportunity costs for the health industry and artificial intelligence engineers (AIE). There is a shortage of studies using quantitative models to test the barrier interrelationship and its effect on AI adoption, especially from the perspective of a developing country like India. Therefore, this study explores the barriers to adopting AI in healthcare in India. A total of 250 barriers related to technology adoption are determined after thoroughly analyzing previous studies and several focus group discussions (FGDs). Barriers are reduced to 16 most relevant barriers through multiple Healthcare expert FGDs and the DELPHI method. Interpretive structural modelling (ISM) and crossimpact matrix multiplication applied to classification (MICMAC) are the analytical techniques used to classify the barriers into different impact levels and importance. The derived outcomes from the ISM and MICMAC methods illustrate that the unavailability of infrastructure and policy support and AI's potential cybersecurity vulnerabilities are the predominant problems for AI adoption in healthcare. The outcome is proposed to add to the theory of technology adoption, which facilitates the organizations in selecting the technology and opening the path for its spread and use by healthcare stakeholders.**

***Index Terms*—Artificial intelligence (AI), barrier, DELPHI, healthcare, interpretive structural modeling (ISM)–crossimpact matrix multiplication applied to classification (MICMAC).**

I. Introduction

THE 21st-century market mechanism is mainly driven by new age technologies (NATs), namely artificial intelligence (AI), business analytics, cyber security, cloud computing, data science, DevOps, UI/UX, full stack web development, metaverse, product management, and salesforce [82]. AI technologies enticed most researchers, scientists, engineers, and technological philosophers, and acted as prudent factors in all human life. For instance, in household applications, AI technologies (home robots, AI television, and smart home devices), smart locks, smart thermostats and meters, AI cooking, and virtual reality devise an effortless life for us. In education (through personalized learning, task automation, smart content creation, and inclusive and universal access), AI-enabled technologies transform the pedagogy and learning ecosystem, shape Industry 6.0 by creating the space for robo-medics, multidimensional printing, alternate energy, deep dive EEG [27], and mark the dawn of a new era in governance and politics (by automating routine tasks, advancing R&D, ensuring security, and composing sound administration). Human intelligence has a few flaws [74], famously explained in "Moravec's Paradox," whereas AI can outsmart human intelligence at many levels (e.g., objective arithmetic operations or computational complexity).

AI possesses a huge potential to derive the global economy through computer vision, natural language, virtual assistants, robotics process automation, advanced ML for image recognition, text generation and analysis, speech recognition, data analytics, and algorithmic recommendation [1]. Also, AI will largely benefit humankind by automating processes, reducing human interaction, improving work quality, facilitating efficient data handling, and widening its range of applications in numerous industries. Human–AI collaboration will help form an effective team in the organization for improved decision-making [99]. Furthermore, the impact of AI can be largely affected by few factors, such as global connectedness, the pace of adoption of AI, and a country's labor market structure [16]. There is a lot of research that shows how important it is for AI and IT to work together to create a sustainable market environment and increase consumption in many fields, including agriculture [8], [46], construction [86], energy [2], manufacturing [3], [61], [66], healthcare [44], [45], logistics and supply chain [8], [76], and education [65].

AI and blockchain technologies are now emerging as powerful drivers of innovation in various sectors, including healthcare. These technologies can potentially revolutionize healthcare delivery by enabling data-driven decision-making, improving patient outcomes, enhancing privacy and security, and optimizing resource allocation [109]. AI adoption is transforming many industries, but its impact on the healthcare sector is life-changing. Recent studies and tests show that AI can deliver identical or better prognoses, diagnoses, and surgical outcomes compared with medical professionals [21], [25], [42], [106]. In the pandemic scenario (mostly in any infectious disease), apart

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The authors are with the Management Development Institute Gurgaon, Gurgaon 122007, India (e-mail: amitkgupta@mdi.ac.in, amitg56@gmail.com; mks@mdi.ac.in).

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from those who get infected, other people's lives are also at risk, whether they are treating them (such as clinicians, nurses, and staff) or coming into contact with them. Now, the question is how AI can be a helping hand in this scenario. Our assertions and recent scientific research illustrate that AI-enabled robots, humanoid robots, and robotic tools can prevent these lives during this kind of pandemic crisis [31].

Moreover, AI-enabled solutions can play a much more intense role in shaping the healthcare industry and improving patients' well-being. The knowledge bricks on "how AI has disrupted (or can disrupt) the healthcare industries and service delivery" became essential in absolute terms. AI has the potential to disrupt the healthcare sector in developed nations and the potential to drive advanced innovations in underdeveloped communities in global health. AI technologies are poised to disrupt our lives (more of a world) with intelligent machines, which enable remarkable cognitive processes, such as perceiving, thinking, learning, decision-making, and problem-solving coupled with advanced data collection, analytics, aggregation, and computer processing power. Disruptive innovations and medical technologies have revolutionized how healthcare is delivered today in India. Prior observations indicate a 15% CAGR growth in the healthcare industry since 2011, estimated to surpass USD 280 billion by 2020. However, providing access to quality healthcare for a population of 1.3 billion is a huge challenge that the country has to deal with. AI can help in reducing annual healthcare expenditure, early disease detection, real-time chronic disease information, and limiting healthcare professionals' exposure to contagious environments [90].

A. Application of AI in Healthcare

1) AI in Treatment or Diagnosis: AI technologies assist researchers in developing diagnostic tools, treatment acceleration [21], [69], early detection (through monitoring genetic mutations and aberrant protein interactions), and diagnosis of cancer [42]. Also, these AI-enabled solutions benefit physicians and clinicians by detecting radiographic abnormalities [77], screening, contact tracing, and containing pandemic or epidemic outbreaks [21], [75]. Machine vision is a powerful tool for diagnosis and surgery [13]. Further, AI supports healthcare professionals in decision-making for diagnosis [95], treatment, and prognosis [40]. Moreover, as AI facilitates infectious disease surveillance, radiological and histopathological image analysis become much more efficient [85]. Surgical navigation systems, intelligent image analysis, and aided diagnosis further contribute to developing healthcare infrastructure in any nation [78]. The AI application was also used for pathology and remote diagnosis [28]. Prior literature also proclaims that the existing AI/ML/DL-based healthcare system delivers better healthcare services and diagnoses than human doctors [106]. AI-enabled software as medical tools (SaMDs) have proven effective for diagnosing diabetes and effectively warning patients in advance in case of stroke [6]. A system-dynamic approach to the employment of digital technology in e-health, particularly in the intensive care unit, can aid in the handling of complicated data and facilitate decision-making [25].

2) AI in Nursing: In Japan, the AI-powered humanoid/robot, namely "cloud ginger," was first used to assist healthcare professionals in delivering medicines and food to the patients and afterward used for entertaining the patients during quarantine [21]. On the other hand, AI-supported voice and facial recognition software reduces nurses' dependence on keyboards. Also, AI-powered virtual assistance facilitates clinicians and nurses using DeepMind technologies. The integration of robots in nursing care activities (e.g., walking upstairs, connecting specialists and patients, pick-up and delivery of supplies, companionship to the elderly in senior homes, and patient hair washing) through robotics and machine vision is something a patient can experience in the existing healthcare sector [18]. Apart from these, AI in healthcare settings can help hospitals achieve a nurse-to-patient ratio [26]. It is asserted that if the safety and assurance approach to AI–healthcare systems is followed, it can ultimately expand the technology-centric focus among the staff and patients in hospitals [97].

3) AI in Healthcare Decision-Making: Previous articles observed that the AI-enabled system could detect and supply the necessary information on infectious disease outbreaks, which can ultimately direct healthcare workers and policymakers to make decisive decisions during epidemics and pandemics [21]. The decision-making capacities of AI technologies were closely observed in the previous literature [88], and it was suggested that if the sector-specific law had been revised and eliminated, discrimination and product liability could deal with the challenges faced by AI technologies during decision-making. An experimental study [64] on consumer receptiveness to AI-based algorithms and statistical models pinpoints that, instead of replacing humans, it instead supports the decisions made by healthcare professionals. Also, the prior articles shed light on the significance of values in motion design to help LGBTIQ+ elders cope with loneliness and depression through value-driven decisions made by the value-sensing robots [83]. Martins et al. [68] explored the implications of AI-based methods in dental prosthesis sectors where the significance of scientific and technological advances seems to define lines of action and decision-making processes clearly. Earlier research [79] explicitly introduced the FairLens methodology, which shows the fictional commercial black-box model can act as a clinical decision-support system based on historical data on patients. Marotta [67] emphasized that the algorithms (free from bias) make medical decisions more efficient by eliminating gender biases from the data. In addition, they propose the idea that network data evolution analysis models and deep-learning approaches can assist management in making better decisions in the healthcare supply chain [9]. Also, AI-driven solutions reduce patient waiting times, increase patient involvement, and control hospital mismanagement [62]. The roaring debate on the robustness and qualification of AI technologies in healthcare was fairly substantiated in the previous cases [4], [21]. It can improve the ability of internal hospital procedures to process information [12]. The research outcomes from medical studies emphasize that AI-driven clinical decision-support tools can provide important insights into RCTs [6], assist medical physicians [42], and enhance clinical decision-making [40], [85].

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4) AI in Medical Devices (Instruments and Equipment): Agree or not, we are all heading toward new digital innovations, the Internet of Things (IoTs), and AI [104], for example, wearable medical devices. In the 21st century, it is a consequential truth that most consumer and patient devices are now often equipped with machine learning (ML) to interpret and evaluate data for clinicians, physicians, firms, consumers, or patients [71]. AI-enabled technologies are now the gist of medical devices in the form of varied medical instruments and equipment. For example, wearable healthcare robotic arms in Japan help physically impaired patients with mobility. Similarly, automation with inherent AI is instantly emerging in medical assistance devices [39], such as SaMDs [6], [73]. Benjamens et al. [11] investigated and validated FDA-approved medical devices and algorithms to raise awareness about these devices (applications) and the regulatory forces surrounding this industry. In healthcare settings, AI-enabled devices (or algorithms) assist clinicians, physicians, and nurses in terms of surgeries (robotics), detection and diagnosis of diseases [6], patient care and elder care, and early warning signs of epidemics and pandemics [19], [21].

B. Research Background

The adoption rate of AI has increased in various industries, and prior articles show that all sectors are adopting AI with huge investments. However, the adoption of AI in the healthcare sector, particularly in India, is not well covered in the literature. India is still struggling to provide primary healthcare to its population. According to industry research reports, AI will significantly disrupt the healthcare industry with interesting results, including applications in "wellness, early detection, diagnosis, decision-making, treatment, end-of-life care, research, and training." According to a McKinsey Global Institute report, AI-enabled healthcare services in the USA may save around 0.7% of GDP, or $300 billion annually, using ML in population health programs. It is projected that the personalization of medications and treatments would save between $2 and $10 trillion [15]. Technology behemoths like Apple, Google, and IBM have all made large investments in healthcare analytics, demonstrating the growing interest in AI technology [54]. Healthcare lags behind other industries in the adoption of AI, despite apparent public preparedness and encouraging outcomes [15]. Bughin et al. [15] observed that between 10% and 20% of AI's potential has been realized in the healthcare industry. The top three obstacles to adoption, according to C-suite executives in the healthcare industry, are technological infrastructure; organizational structure and procedures; and strategy, leadership, and talent.

C. Research Objective and Questions

Despite the growing interest in AI and blockchain technologies in healthcare, there is limited research that comprehensively examines the regulatory landscape for their implementation. Integrating AI and blockchain technologies in healthcare also raises many concerns related to data privacy, security, ethical use, technological challenges, and potential negative consequences for different stakeholders. Thus, we intend to first detect the major barriers to AI adoption in developing nations, India by reviewing the literature methodically; second, we identify the association among the various barriers to AI adoption and compose their hierarchal structure model using interpretive structural modeling (ISM); and at last, we determine the dependence and driving power of these challenges through crossimpact matrix multiplication applied to classification (MICMAC) methodology [48]. This article will assist concerned authorities in recognizing barriers and advancing policies to tackle them effectively. The literature used the theory of technology adoption and models like TAM and TOE to adopt e-banking [51] and online impulse-buying behavior [50]. Similarly, the literature used the social cognition theory within the framework of human–technology interaction for using e-banking in a transition economy [53]. Thus, the article's outcome is proposed to add to the technology adoption theory, which facilitates organizations in selecting the technology [63] and opening the path for spread and use by healthcare stakeholders [24]. The article proposed to address the following research questions.

RQ1: Which barriers impede AI adoption in the healthcare sector?

RQ2: What are the interrelationships and hierarchical models that exist between the most relevant barriers impeding the adoption of AI in the healthcare sector in India?

RQ3: What are the most prominent (driver) barriers to adopting AI in healthcare in India?

II. Literature Review

A. Literature Search Criteria and Database

The database utilized is Scopus and Google Scholar. We used Boolean search using "Title and Abstract" *(TITLE-ABS()) in Scopus.* Inclusion criteria include English language, source type as a journal, document type as an Article, subject area as "Business Management and Accounting," "Decision Science," "Engineering," and "Medicine," and Time period: 2010 to 2023. More than 95 articles are referenced in the article.

B. Barriers to the Adoption of AI in Different Organizations

Access to information in a digitized organization and managers operating outside the relevant flow of information lead to administrative disruptions. Adding more, ineffective or local search routines, unassertive data, workforce skills, the nexus between technology and humans, and the convertibility of information into actionable form are the major causes of concerns in the formative technological business environment [37]. Further, the obvious challenges, such as the availability of Big Data, hidden assumptions in data, reliability of information-in-hand, and its optimum utilization, are the main reasons that favor the human workforce over AI [19], [37], [59]. The contentious issues of robots' presence around us and conscious thoughts that they are linked to the internet bring fear of hacks, information stealth, deterrence, and distrust toward robots and AI. Additionally, the sense of security, connections, friendly service provisions, and acceptance were among a few other factors that could explain the contrast between LGBTIQ elderly care and robots [83].

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Further, the sensitive information that contains gender identity, sexual orientation, and intimate and granular data can cause a sense of distrust in AI if it is shared with marketers who have no respect for individual privacy and dignity or in the hands of hijackers who can use these data to fuel offensive jargon and communal rage [28], [83], [87]. The user's safety, secured surroundings, emergency aid, and social desirability of AI-enabled solutions can be disheartened if safety and security are not prioritized [28], [83]. In addition, the prominent human workforce, reliability limitations, and narrow AI capabilities restrict the general public from adopting these AI-enabled services [19]. Due to AI's infancy, industrialists and various stakeholders have severe doubts about the suitability of automation for identity-based consumption. Also, the suspicion of the effectiveness of an algorithm's recommendation, human instinct (trusting one's judgment), lack of emotions, and involved risk can be major barriers to AI adoption [107]. Through detailed discussion and expert opinions from a multidisciplinary perspective, the researchers concluded that challenges must be addressed to flourish and cherish these NATs (especially AI). These challenges range from social fear (from technology, implicit bias, human moral principles, and hacker attack) to economic fears (costly affairs, gains and losses, and job losses), to technological fear (volume, variety, velocity of data, ML for imaging context, human vs. computer interaction, information flow, interpretation, and autonomous intelligence), to political and regulatory fear (lack of transparency, reproducibility, copyright issues, accountability, and lack of trust) among the various stakeholders in the production–consumption web [28].

Further, AI-enabled technology, services, and solutions ignore socially relevant and inclusive operations management. Due to its expense, inequitable access to resources, and technological disparities in the world demographics, for example, developed vs. least developed nations [98]. After analyzing 88 research papers on employee–robot interactions in service industries [23], the predominant impediments, such as effortless collaboration (between humans and robots, as they called "cobots"), hyped-up robotic potential, intricacy in interdisciplinary research (robotics, computer sciences, and engineering), and standard learnings were emphasized more in prior literature. Additionally, these AI-enabled technologies intensify the inequalities within society (by disrupting physical work through automation), abate judicial accountability (for the development and implementation of AI-enabled systems), espouse unrestrained regulatory power (which can cause damage in the wrong hands), and foster obnoxious mass surveillance to improve its functional mechanism [87].

C. Barrier to the AI Adoption in Healthcare

If we explore to the full extent, we observe the diverse setbacks that act as stumbling blocks for adopting AI in healthcare in discrete demographic settings. The scarcity of innovative aptness and frail leadership to drive technology in healthcare habitats can be crucial for adoption and technological convergence [26]. Also, the agitations toward new intelligent products and the growing fear that specialists may be replaced by machines sooner or later are temporizing the adoption of AI in healthcare. In addition, the suspicion of an unappealing transformation in clinical duties causes major distress among healthcare workers, which ultimately appears as resistance toward AI-enabled technologies [18], [78], [104]. Further, the problem of free-ride algorithms, a sense of accountability in incidents under autonomous systems, taking back control from automated systems, and required standards for AI adoption are still ambiguous for the healthcare sector [97]. This further calls into question the responsibility of safe patient care, which clinicians believe they can handle better than an automated system. Also, the looming questions, such as whether it is better to trust machines in terms of rationality or whether we can trust AI or machines emotionally, how can a robot or AI-enabled technology emotionally bond with patients? [55], [57], [97]. Also, anxiety ideation and mixed feelings (toward AI, technology, healthcare delivery, diagnosis aftereffect, surveillance, and delegation) erect a sense of skepticism within society, eventually intimidating AI adoption in healthcare sectors [97]. Moreover, stakeholders' major concern was how these AI-enabled technologies could maintain sound interactions (or triangular relationships) between clinicians, patients, and their relatives [97], [106]. Due to the absence of local models' heterogeneity, it limits the global model architecture for personalized monitoring [92]. The AI-enabled technology or ML massively depends on the data used to train the algorithm and if the data reflect a bias toward the minority (due to data disparity, irrelevant populations, nonrepresentative specifics, and inappropriate sampling), which ultimately creates statistical prejudice and impacts the derived outcome from the analysis. It can mislead the medical professional by disclosing biased visualization labels and models (through selection and spectrum bias, interpretation bias, poor quality of data, and insufficient data) by generating a dodgy algorithm for AI-model development [14], [22], [40], [67], [69], [70], [85], [89], [95].

The previous articles evidently raise skepticism about the accuracy of these AI-enabled technologies during diagnostic procedures and patient treatment [4], [14], [30], [89], [104]. In terms of diverse patient populations and clinical trials (for example, trials of new drugs), ensuring precision (in terms of safety and fairness) in an ML-based clinical decision-support system is a deliberate and discrete task that includes the whole proceedings from data to action [29], [79], [89]. In the healthcare sector, identifying criteria for service quality reflection falls under "problem structuring." Medical records should precisely match those of the patients, and a minute mismatch may compromise the diagnostic procedure or patient safety [49], [68]. In AI-enabled technologies, the "black box" problem arises due to deep-learning intricacies and sophisticated algorithms, which make the decision-making process rather opaque [85]. Further, patients may feel reluctant to share personal information, or participants may express concern about sharing information about their families or relatives [71], [80]. The lack of security and privacy is a huge blockade among healthcare professionals in the AI adoption process. The information they possess or process is so sensitive that if it is not managed (processed, stored, and distributed) securely or the connection is not protected, it can be exposed to cybersecurity vulnerabilities [4], [12], [33], [49],

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[67], [71], [73], [106]. The prior articles indicate the patients' (users') perspectives, especially elders, family members, and the general public, who are extremely concerned about the probable data breach and privacy invasion [33], [55], [70], [71], [104]. Few articles also pinpoint the efficiency obstacles around AI-enabled solutions (such as AI applications and dynamic storage) designed to manage patient outflows and map the sociotechnical impediments (such as legislation to clarify responsibilities) among the aging populations [62], [104]. It is also a matter of debate and usual dilemma among healthcare providers and patients about whether it is worth investing in these costly technologies (AI automation) and looming doubts about the derived benefits from them because there is limited and dubious information on the after-effects and cost [69], [104]. The conventional financial barriers (such as insufficient funding, administrative costs, and financial sustainability) to funding, irrespective of huge public demand, remain a worry for the public healthcare department due to the immense expenditure involved in acquiring and retaining AI technology [34], [104]. If we delve deep, in the case of AI-backed solutions, research exhibits the lack of trust or perceived trust among the patients or users [33], [57], [69], [70], [79], [96], [104], [106]. Besides, healthcare industries demand personalized experiences (analytical data services, predictions, and data usage for automated decision-making) and tailored treatment based on the nature of the diseases or real-time data [32], [32], [55], [73], [104]. Sometimes, psychological elements (e.g., strange design of devices, social pressure, and collective perceptions) also derive the anti-AI movements due to assorted resistance (psychological, cultural, workforce, and AI paradox) in the course (or early stage) of AI adoption in healthcare [85], [104]. Technological challenges, such as untraceable bias (in the case of black-box ML models), language differentiations (use of informal and nontechnical terms among patients), and slow technology adoption can lead to redundancy during the AI adoption in public healthcare [79], [85], [103]. Also, the dearth of explainable AI in healthcare decision-support systems constructs a sequential blockade for AI acquisition and retention [70], [79], [106]. The AI and ML models need to be more transparent and competitive enough to develop error-free inferences for decision-making. These models should also address possible bias and incidental consequences [40], [79], [102]. AI technologies require a reliable network and consistent performance of AI (for example, in radiology), which is quite stodgy in the infancy stage of these technologies [17], [96]. With all that being said, the participant's or patient's willingness to bear the danger of unfavorable health consequences and the precarious nature of medical devices based on AI or ML algorithms exacerbate the healthcare discrepancy [11], [69], [80]. The ethical issues in AI adoptions were widely spotlighted in prior literature, which comprises various elements, such as emulations of feelings, ethical values (trust, transparency, accessibility, and social well-being), ethical implications (medical monitoring), dilemmas between humans and AI, and disease diagnoses in the absence of assured treatment, to name a few [4], [13], [14], [34], [90], [91], [106]. Who will own the data and who is going to protect (or regulate) these massive amounts of data is also a cause of concern among the various stakeholders in the healthcare sector [14]. The vendors (technology), research, education, and expected return on investment add to the interpretability barriers; and postworkforce displacement, regulators, and stakeholder engagement amplify the control disputes [4].

In addition, the efficacy of AI/ML/DL algorithms largely depends on the inputs [in terms of data, context, utility, confidentiality level, size, availability, paucity, language processing systems (or NLP), structured data, and quantity of medical data] and its significance for diagnosis and treatment. With that, how the data are regulated (usage and scarcity of data) and governed (prohibited from using data) by the authorities determines the technology adoption competency [30], [32], [34], [42], [62], [72], [73], [85], [91], [95], [102]. There may be a difference between the national policies on AI, international standardization on safety obligations on robotic devices or robots, and skepticism on the commercialization of AI-enabled technologies, which also acts as a barrier to AI adoption [41]. And, patients' skepticism toward the service capabilities (quick, effective, conditions, and need specifications) of these AI-fueled intelligent machines retreats the adoption attitude among the public [55]. The result derived from Big Data analytics and AI technologies (BDA–AI) equally encounters subjectivity and uncertainty issues in medical decision-making, which leads to redundant tasks for medical professionals [12]. In the editorial, Angus [6] asserts that the magnitude of the contribution made by the SaMDs to reduce hypotension is hazy, measuring the intensity of AI-enabled device deployment is demanding, and questions on assessment of healthcare interventions and discharge are still open to debate. Apart from that, education and learning became quite complex in the AI-backed healthcare ecosystem due to the medical curriculum, the integration of doctor–engineering courses, and rigorous clinical intelligence [6], [14]. Unsupervised learning in AI is still in the development stage and needs more enhancement to eradicate the errors in the algorithm. Also, the diseases need to be classified based on their nature, as do the clinical experiments and interfaces between humans and computers [42]. The dearth of replication on clinical efficiency (if we compare AI to humans) due to unreliable design, algorithm validation, reflective designs, and dehumanization of medicine (AI replacement) are the major concerns hovering around healthcare professionals [14]. The crucial role of algorithmic stewardship, creation and maintenance of algorithm inventory, constructing logical and error-free algorithms, algorithm manipulation in clinical decision-support systems, and administration of sophisticated algorithms is a challenging and delicate task [29], [30], [49], [85]. The emerging AI-enabled technologies are believed to adversely affect drug development, time, and resource consumption in the healthcare sector [11], [29]. Additionally, there will always be the possibility of unknown and unintended repercussions of AI and ML due to the inherent risks associated with these medical devices [11], [69]. The awareness of regulatory bodies, the dire need for autonomous regulators for AI interventions, the lack of regulations and legislation, and the distortion of harmony between regulatory channels discourage AI development and

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TABLE I
TABLE SUMMARIZING THE DEMOGRAPHIC DETAILS OF THE HEALTHCARE EXPERTS

DELPHI stage experts			
	Experts	Percentage	Av. ex. (years)
Position/designation	Doctors	26.7%	18
	Nurses	16.7%	15
	Senior lab experts	40.0%	14
	Adminstration	16.7%	17
Gender	Male	56.50%	20
	Female	43.50%	16
ISM-MICMAC stage experts			
Experts	Numbers	Position/ designation	Av. exp.
Doctors	2	Senior consultant/consultant	17 years
Professor	2	Senior Professor (Gyne/Cardio)	18 years
Chief Doctors	1	HOD	20 years
Nurse	2	Head nurse	24 years
Heath Department Officer	1	Deputy Director	25 years

adoption in hospitals [11], [69], [85], [90]. In addition, the stringent (rigorous) requirements imposed by regulators for medical device licensing (to supply in the market) create a huge void between suppliers and service providers [11]. The prior articles raise issues, such as: what if deep learning overlooks a diagnosis, or who will be responsible if medical treatment goes sideways? The consequences of these errors can adversely affect patient health and well-being [70], [85]. The minimal participation of patients when designing the algorithm for decision-making and cyber security vulnerabilities can disrupt the patient care and treatment process across healthcare facilities [70], [73]. Also, augmented intelligence for clinical decision-support systems or medical purposes is considered primitive [70]. The limited priorities given to equity and inclusion in healthcare settings are still not fully explored by marketers, researchers, and regulators [34], [69]. Furthermore, Guo et al. [34] and Schwalbe and Wahl [90] highlight the significant strand of practical affairs of AI adoption and implementation, such as research potential, infrastructure-based development, regional disparities (between least developed and developed nations), and practical considerations of AI grounded on lab-based AI intervention. The more exhaustive treasure trove of barriers to AI adoption in healthcare is summarized in Table I.

D. Different Techniques Used in the Research Related to Barriers, Issues, Challenges, Drivers, etc., in the Adoption of AI in Different Sectors

Our exploration of the methodological vogue exercised in prior article depicts that the studies on barriers to AI in healthcare industries proportionately diverged in qualitative, quantitative, conceptual, analytical frameworks, and mixed approaches. Much of the qualitative research framework adopted the interview schedule, thematic analysis, and section analysis to depict the prevailing setbacks for AI adoption in healthcare industries. The authors in [97] and [102] tried to postulate safety, assurance, and contextual integrity in the U.K. concerning AI in healthcare, employing thematic and case studies. However, the studies conducted in the USA and Europe explicitly comprise the contextual, legal, and directional hindrances through conceptual structure, systematic review, and the NASS framework [26], [39], [89], [96]. Whereas, in Asia Pacific (Australia, China, India, and Japan), the qualitative, thematic, and sectional investigation led the researchers to explore regulatory barriers, knowledge embodiment, sociotechnical impediments, and barriers to innovation for AI adoption in the healthcare industries [41], [55], [81], [83], [104]. Further, the exploratory research and case-study-based approach signifies the role of AI in elevating smart health and value cocreation in healthcare [60], [106].

Research methods, such as systematic literature reviews, cluster analysis, and conceptual propositions, are crucial for research progress in any domain. The studies by Dwivedi et al. [28] and Haefner et al. [37] set the research agenda for the implication of AI in multidisciplinary (not specific to healthcare only) post-SLR. Whereas, few articles [19], [71], [90], [91] anticipate the future article agenda for AI in healthcare domains (cost disease, intelligent automation, and AI-enabled clinical solutions) after performing systematic reviews of past articles. Further, some researchers' [59], [67], [89], [98] intent to conduct systematic reviews of past articles is essential to specify the ethical and legal implications, standard liabilities, social inclusion, and economic and relational consumer value of AI in healthcare settings. Moreover, Briganti & Moine [14], Ellahham [30], Mohamadou et al. [72], Naseem et al. [75], Sipior [95], and Yang et al. [106] absorbed the implications and AI barriers in healthcare in terms of SmartHealth, healthcare responsiveness, AI in medicine, predictive and management capacity, and chronic disease care. Other than that, thematic analysis was used to analyze safety and assurance, value sensing, sociotechnical barriers, customer-relationship capabilities, and sustainable development in healthcare [20], [55], [83], [97], [104]. Further, section analysis [81] and content analysis [33] were applied by researchers to identify the various forms of knowledge embodiments, benefits, and challenges of m-health. In addition, De Keyser and Kunz et al. [23] and Latinovic and Chatterjee [59] employed the mapping model and TCCM framework to synthesize their vast understanding of previous articles for future article endeavors. The PRISMA extension for scoping reviews [75] and bibliometric analysis [34], [91] are also exploited by researchers to indicate emerging themes in AI and healthcare research. At least, we would like to emphasize the significance of conceptual research because, at the industrial infancy phase (AI at present), AI in healthcare can botch into the theoretical mass without appropriate layout and research directions. In a conclusive sense, the authors in [19], [26], [71], and [108] draw the conceptual layout for future implications of AI or AI-enabled solutions in healthcare sectors, whereas Dwivedi et al. [28] indicate the broader application of AI in various industrial settings through conceptual underpinning. Moreover, the research concentrations also attracted the ethical implications, liabilities, algorithmic stewardship, data regulations, reporting standards, social inclusiveness, and customer values of AI-enabled solutions in the healthcare sector while instituting their conceptual layout [29], [33], [40], [49], [59], [67], [73], [87], [98]. Further, Sipior [95] proposed developing

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and appropriately using AI in the wake (or in response) to the COVID crisis. Few articles specify the conceptual development for AI adoption in healthcare for certain diagnoses, such as cancer [42], diabetes [30], and medical practices [13], such as medical imaging [77]. The conceptual papers by Clancy [18] and Meskó and Görög [70] demonstrate the significance and challenges of AI for medical professionals and nurses. Finally, Matheny et al. [69] and Reddy et al. [85] indicate the benefits and challenges of AI-enabled healthcare delivery.

As per our literary investigation, most researchers exploited the quantitative methods through lab experiments, clinical trials, questionnaire surveys, and advanced data analytics (AI, ML, and DL). The experiment/clinical trial conducted by Angus [6] and Park et al. [80] used descriptive analysis, hierarchical regression, and a structural model to examine the competing effects (of emotions). The cause-and-effect or linkages between AI adoption and nonadoption in the healthcare sector were further assessed in various articles using basic and advanced regression techniques to predict AI accuracy in the Taiwanese healthcare system and AI medical resistance in the USA [64], [105]. Support vector machine regression, model-tree-based regression, and multiple linear regression analysis [17] to estimate the healthcare expenditure in Turkey; IBCC and GP regression [32] to examine the impact of automation in healthcare in pursuit of a better healthcare system in Europe and North America; and logistic regression [72] to predict and create a better healthcare system during the COVID crisis. Also, the studies from China, France, and South Africa investigated the behavioral intentions, healthcare problems, BDA in the healthcare supply chain, and absorptive capacity of AI by using CFA and SEM or PLS–SEM [10], [12], [55], [78]. The *t*-test and ANOVA were used in earlier articles [57], [64] to measure the differences and variances between the groups (or more) for human resistance, patient response, and cognitive and affective trust for AI in preventive care operations confined to South Korea and the USA. Research also uses theoretical frameworks and models to adopt technology like UTAT [52].

Moreover, with the exponential growth in computational science, cyberspace, virtual consumerism, and Big Data availability in velocity, volume, and variety, various researchers sought ML, deep learning, reinforcement learning techniques, and AI to act upon or make clinical decisions based on real-time data. The different mathematical and algorithmic tools were developed by various researchers, such as artificial neural networks [17], [22], [67], [92] for personalized activity monitoring, estimation of healthcare expenditure, addressing liabilities, and pandemic prevention. Likewise, prior articles examined the barriers to AI adoption in healthcare using multilayer perceptron deep artificial neural networks [22], convolutional neural networks [72], [92], bidirectional long short-term memory [92], hierarchical Bayesian module [64], strategic options development and analysis [68], cognitive mapping [68], and measuring attractiveness by a categorically based evaluation technique [68]. Yun et al. [107] differentiate patients' behavioral and neural responses toward doctors and medical AI through fMRI, region-of-interest analysis, and a general linear model in South Korea. Further, Panigutti et al. [79] used the Fairlens analysis and sequential ML model to audit the clinical decision support systems. Ceylan and Atalan [17] used genetic-algorithm-based feature selection, relevance vector machines, and random forests [22], [72] for the estimation of healthcare expenditure and pandemic prevention during the COVID healthcare crisis. Prior article [67] using forward–backward analysis, fuzzy logic, Bayesian networks, and general ML techniques draws attention to the liabilities of AI technologies, especially in the women's healthcare system. The network data envelopment analysis, deep-learning genetic algorithm, ML, and Shapley additive explanations [9], [22], [103] were used to forecast the sustainability of healthcare supply chains in Iran and preventive care (for example, drug event detection, extraction, and pandemic prevention) in the USA. The queuing model simulation, *M/M/k* queuing system model, and shortest consultation time model demonstrate the effectiveness of AI in enabling solutions for managing outpatient flow [62]. In addition, Kumar et al. [56] explored the AI-enabled CRM potential employing the correlated topic model, latent Dirichlet allocation, structural topic model, and natural language processing in Indian healthcare. The historical awareness multilevel embedding model [103] measured the efficiency of the historical profile in drug event detection and extraction. In prior article, Bayesian ML [32], post-hoc analysis, and predictive ML modeling [10] were adopted for organizational and healthcare decision-making. Also, various other mathematical and algorithmic models, such as the susceptible–infected–recovered model, the susceptible–exposed–infected–removed model, the susceptible-infectious quarantined–recovered model, the susceptible–exposed–symptomatic–asymptomatic–recovered–seafood market model, logistic regression, support vector machines, naive Bayes, decision trees, linear discriminant analysis, U-net, unsupervised learning [72], spread, sustainability framework, nonadoption, abandonment, and scaleup [96] were utilized to understand the AI adoption challenges in healthcare sectors.

Though limited, multicriteria decision-making (MCDM) tools are also used in the article. Badi et al. [4] and Damoah et al. [20] used the thematic mapping, inductive approach, and AHP method to examine the challenges for AI in healthcare and the healthcare supply chain in Ghana and the UAE. In addition, Badi et al. [4] and Longoni et al. [64] used correlations and AHP to find the associative links between human resistance to medical AI, challenges in decision modeling, and AI adoption in the USA and UAE healthcare sectors.

III. Methodology

The study used DELPHI and ISM–MICMAC to analyze and evaluate the research question framed. DELPHI techniques help to reduce the barriers identified from literature from 250 to 21, and finally, to relevant 16 barriers to AI adoption in the healthcare industry in India. ISM–MICMAC groups the barriers into dependent, independent, and affinities based on the behavioral barriers when interconnected. Finally, the most important barriers (based on driving power) are identified. The flow chart for research methods is shown in Fig. 1.

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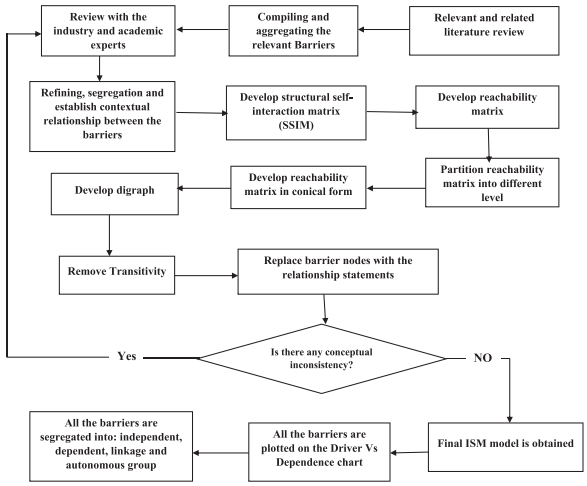

Fig. 1. Figure represents the flow chart for the research method followed in the research (taken from [35]).

TABLE II
TABLE SUMMARIZING THE BARRIERS AS AN OUTCOME OF THE DELPHI TECHNIQUE

S. No.	Barriers	Mean	Count	COV
1	Unavailability of infrastructure and policy support to AI	4.71	30	0.16
2	Government support, regulations,and funding	4.43	30	0.18
3	Limited use case of AI applications in healthcare	3.91	30	0.17
4	Transparency and fairness	3.91	30	0.17
5	Lack of interoperability is a huge limiting factor for AI	4.26	30	0.18
6	AI's potential cybersecurity vulnerabilities	4.48	30	0.19
7	Data availability and quality challenges	4.48	30	0.16
8	Lack of trust in AI	4.22	30	0.19
9	High implementation cost	4.39	30	0.18
10	Shortage of AI talent causes more delays in adoption	3.91	30	0.18
11	Resistance to change	3.91	30	0.18
12	Lack of clear strategy	4.26	30	0.18
13	Limited knowledge and understanding of AI	3.96	30	0.19
14	Lack of leaders' ownership of and commitment to AI	4.26	30	0.18
15	Performance and accuracy of AI-enabled technology	3.91	30	0.19
16	Job security and machine — human interactions	4.35	30	0.19
17	Fear of ineffective AI implementation	3.43	30	0.3
18	Ethical issues	3.52	30	0.22
19	Analytical and outcome biases	3.74	30	0.24
20	An incomplete digital platform	3.74	30	0.24
21	Emotional constraints	3.52	30	0.24

A. Identification of Barriers to Adoption of AI in Healthcare: Outcome of Literature Review and DELPHI Technique

We identified and accumulated around 250 barriers (Table I) from different studies based on the adoption of AI. These studies were conducted in different sectors, demographics, or geographies. Most of the barriers identified had similar meanings or reflected similar barriers; thus, the similar barriers were clubbed together with the help of academic and healthcare experts. Finally, we have 21 unique barriers. A structured questionnaire was designed to rate these barriers on a scale of importance (a response point of 5, where 1 stands for not-important and 5 is important) in healthcare. These questionnaires were circulated to 30 healthcare experts (doctors, senior lab experts, nurses, administrators, radiologists, pathologists, and suppliers) representing top hospitals in India. These experts had more than ten years of experience in the healthcare industry. The demographic details are summarized in Table I.

The response received is analyzed using the DELPHI method to identify the relevant barriers. A combination of the mean (>3.5) of the response and the coefficient of variance (COV 0.2) for each barrier is used to identify the barriers [93] in Table II.

B. Interrelationships and Hierarchical Models That Exist Between the Pertinent Barriers That Impede the AI Adoption in Indian Healthcare: Application of ISM–MICMAC

We explore the prevalent association between the relevant identified barriers and establish the hierarchical relationship between them using ISM–MICMAC, an MCDM technique proposed by Warfield [101]. ISM–MICMAC is a multidimensional technique based on area experts' qualitative judgment and experience to segment the barrior into different groups: dependent, independent, autonomous, and probable linkages. These segregations are based on behavior barriers when interacted with or interconnected [43].

1) Steps in ISM Technique: The famous ISM method was adopted over other methods for various reasons. First and foremost, there are multitudes of blockades looking for a way out that is complex in nature and difficult to understand because there are multiple nexuses between the variables. Second, it helps to determine these associations and interdependencies with expert help, whereas SEM is a statistical technique that only provides factors effecting a problem. However, SEM fails where there are interrelationships among the variables [48]. Third, the ISM technique is a group exercise that turns the undistinguishable and abysmally complex structures into a well-defined model [7]. Fourth, the redundant research culture to refer to prior articles is not required ISM application, and last, ISM can be practiced in numerous industries, such as education, healthcare, supply chain manufacturing, etc. [38]. Various researchers and practitioners excised the ISM model to demonstrate the order and composite association amid the variables [94], the figurative expression of the variables in structural form, and successfully adapted to analyze the barriers to AI adoption in India's engineering industries. Anantatmula and Kanungo [5] utilized the ISM–MICMAC method in business organizations to effectively analyze the barriers to adopting AI. They applied the ISM and MICMAC techniques to rural healthcare in India to detect and discern the barriers to AI growth. Karamat et al. [48] used the technique for the same prospect in a different context, such as barriers in knowledge management in Pakistan from healthcare perspectives and policies. The ISM application and use in research articles are summarized in Table III.

The article follows the steps involved in the techniques outlined as follows [35], [47].

1) Identify the barriers to AI adoption through an exhaustive literature review.
2) List the barriers by exploiting expert opinion and extensive brainstorming.
3) Develop a structural self-interaction matrix to exhibit pairwise linkages between the variables.
4) Develop an initial reachability matrix by binary digitizing the structural self-interaction matrix with 0 and 1.

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TABLE III
TABLE COMPILES THE SUMMARY OF ISM–MICMAC USED IN DIFFERENT FIELD OF STUDY

Author and year	Technique used	Area of study
Gorane and Kant (2015)	ISM, fuzzy and MICMAC	Supply chain management
Digalwar and Giridhar (2015)	ISM	Electric mobility
Dewangan et al. (2015, 2017)	ISM, fuzzy and MICMAC	Competitiveness in the Indian manufacturing sector
Sindhu et al. (2016)	ISM, fuzzy and MICMAC	Solar energy
Attri and Grover (2017)	ISM and MICMAC	Designing the product life cycle
Mangla et al. (2018)	ISM– fuzzy DEMATEL	Agri-food supply chains
Kannan (2018)	DELPHI–ISM–ANP	Sustainable supplier selection process
Zhou et al. (2019)	ISM–MICMAC	End-of-life vehicle (ELV) recycling management
Kaswan and Rathi (2019)	ISM–MICMAC	Green lean six sigma implementation
Rana et al. (2019)	ISM–MICMAC	M-commerce adoption in SMEs in the UK
Ahmad et al. (2019)	ISM and MICMAC	Geology
Moktadir et al. (2019)	ISM and TISM	Human resource
Gupta and Goyal (2021)	ISM_MICMAC-FAHP	BDA in manufacturing in India
Karamat et al. (2018)	ISM-MICMAC	Healthcare–Pakistan
Anantatmula and Kanungo (2014)	ISM-MICMAC	AI in rural Healthcare in India
Raut et al. (2021)	ISM-DEMATEL	Indian manufacturing supply chain

TABLE IV
TABLE SUMMARIZES THE STRUCTURAL SELF-INTERACTION MATRIX FOR BARRIERS OF ARTIFICIAL INTELLIGENCE ADOPTION IN THE HEALTHCARE SECTOR OF INDIA

Code	Barriers	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2
AIA1	Unavailability of infrastructure and policy support to AI	O	V	V	X	X	V	V	V	V	V	V	V	V	V	V
AIA2	Government support, regulations,and funding	O	A	A	A	A	A	V	V	V	X	X	A	A	V	
AIA3	Limited use case of AI applications in healthcare	V	V	A	A	O	A	A	A	A	A	A	O	V		
AIA4	Transparency and fairness	O	V	A	A	O	A	V	A	A	A	A	O			
AIA5	Lack of interoperability is a huge limiting factor for AI	O	O	A	A	A	O	A	A	O	A	V				
AIA6	AI's potential cybersecurity vulnerabilities	O	A	A	A	A	A	V	V	V	O					
AIA7	Data availability and quality challenges	A	A	A	A	O	A	A	V	V						
AIA8	Lack of trust in AI	A	A	A	A	A	A	A	A							
AIA9	High implementation cost	O	A	A	A	A	A	V								
AIA10	Shortage of AI talent causes more delays in adoption	O	A	A	A	A	O									
AIA11	Resistance to change	O	A	A	A	A										
AIA12	Lack of clear strategy	O	A	O	X											
AIA13	Limited knowledge and understanding of AI	O	A	O												
AIA14	Lack of leaders' ownership of and commitment to AI	O	O													
AIA15	Performance and accuracy of AI-enabled technology	O														
AIA16	Job security and machine–human interactions															

5) By using the transitivity method, the concealed connections are explained, e.g., if $X \geq Y$, and $Y \geq Z$, then $X \geq Z$.
6) Final reachability matrix is obtained by eliminating the transitivity.
7) Level partitions are made by giving each variable a different level.
8) Based on level partitions, ISM model is constructed.
9) The MICMAC analysis deployed to determine driving and dependence power. These steps are explained in detail as follows.

a) Structural self-interaction matrix (SSIM): To develop SSIM, experts help to establish (to some extent) contextual relationships between the stated barriers [84]. An expert of eight was requested to give the response. The expert comprises two medical college professors, two doctors, one chief doctor, two head nurses, and one Deputy Director from the Health Department. All the experts have more than 15 years of experience. The details of the experts are summarized in Table I. Response obtained on the 16 barriers are summarized in Table IV. The cell *Aij* depicts the association between *Ai* (*Y*-axis) and *Aj* (*X*-axis). The relationship was annotated as *V*, *A*, *X*, and *O* [35], [47], [48].

1) *V* denotes that barrier *i* influences barrier *j*.

TABLE V
TABLE REFLECTS THE INITIAL REACHABILITY MATRIX FOR BARRIERS TO AI ADOPTION IN THE HEALTHCARE SECTOR OF INDIA

S. No.	Barriers	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	AIA1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
2	AIA2	0	1	1	0	0	1	1	1	1	1	0	0	0	0	0	0
3	AIA3	0	0	1	1	0	0	0	0	0	0	0	0	0	0	1	1
4	AIA4	0	1	0	1	0	0	0	0	0	1	0	0	0	0	1	0
5	AIA5	0	1	0	0	1	1	0	0	0	0	0	0	0	0	0	0
6	AIA6	0	1	1	1	0	1	0	1	1	1	0	0	0	0	0	0
7	AIA7	0	1	1	1	1	0	1	1	1	0	0	0	0	0	0	0
8	AIA8	0	0	1	1	0	0	0	1	0	0	0	0	0	0	0	0
9	AIA9	0	0	1	1	1	0	0	1	1	1	0	0	0	0	0	0
10	AIA10	0	0	1	0	1	0	1	1	0	1	0	0	0	0	0	0
11	AIA11	0	1	1	1	0	1	1	1	1	0	1	0	0	0	0	0
12	AIA12	1	1	0	0	1	1	0	1	1	1	1	1	1	0	0	0
13	AIA13	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0
14	AIA14	0	1	1	1	1	1	1	1	1	1	1	0	0	1	0	0
15	AIA15	0	1	0	0	0	1	1	1	1	1	1	1	1	0	1	0
16	AIA16	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	1

2) *A* denotes that barrier *j* influences barrier *i*.
3) *X* denotes that barrier *i* and barrier *j* influence each other.
4) *O* denotes no influence between barrier *i* and *j*.

b) Initial reachability matrix (IRM): After SSIM, we developed IRM by replacing *V*, *A*, *X*, and *O* symbols with 1 and 0. The conversion into binary format follows a guidelines mentioned in [35], [47], and [48]. The following changes are done in SSIM matrix table.

1) Replace the Cell (*i*, *j*) and Cell (*j*, *i*) with the *V* symbol with 1 and 0.
2) Replace the Cell (*i*, *j*) and Cell (*j*, *i*) with the *A* symbol with 0 and 1.
3) Replace the Cell (*i*, *j*) and Cell (*j*, *i*) with the *X* symbol with 1.
4) Replace the Cell (*i*, *j*) and Cell (*j*, *i*) with the *O* symbol with 0.

By applying these rules, Table V was derived.

c) Final reachability matrix (FRM): The transitivity is established among the drivers in the IRM table to develop the FRM matrix table. The transitivity can be understood as follows: if *X* is related to *Y* and *Y* is related to *Z*, then *X* and *Z* should relate to each other somehow. Transitivity is done to ensure consistency of the expert response on contextual relationships. FRM is obtained by eliminating the inconsistency in IRM by replacing it with 1∗ (Table VI). The sum of 1 in rows gives the driver power of the variables, while adding the 1 columnwise gives dependence power of the variables. The driving and dependence power of barriers in this article is given in Table VI [35], [47], [48].

d) Level partitions: The reachability, antecedent, and interaction sets for each barrier are derived from the FRM. At the intersection of reachability and antecedent sets, the intersection set is created in which variables are taken, which is common in both these sets. The barriers with the same reachability and intersection sets go up the ISM hierarchy. It indicates that the barrier at the highest level in the ISM model will not lead to the barriers above it. Once discovered, the top-level barrier is removed from the remaining variables. For the level partition of these barriers, numerous iterations are developed. Until all the barriers are assigned levels, the stated steps will resume.

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TABLE VI
TABLE SHOWS THE FINAL REACHABILITY MATRIX FOR BARRIERS OF AI ADOPTION IN THE HEALTHCARE SECTOR OF INDIA

S.N	Barrier	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Driving Power
1	AIA1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1*	16
2	AIA2	0	1	1	1*	1*	1	1	1	1	1	0	0	0	0	1*	1*	11
3	AIA3	0	1	1	1	0	1*	1*	1*	1*	1*	1*	1*	1*	0	1	1	13
4	AIA4	0	1	1*	1	1*	1*	1*	1*	1*	1*	1*	1*	1*	0	1	0	13
5	AIA5	0	1	1	1	1	1	1*	1*	1*	1*	0	0	0	0	1*	1*	11
6	AIA6	0	1	1	1	1	1	1*	1	1	1	0	0	0	0	1*	1*	11
7	AIA7	0	1	1	1	1	1	1	1	1	1*	0	0	0	0	1*	1*	11
8	AIA8	0	1*	1	1	0	0	0	1	0	0	0	0	0	1	1*	1*	7
9	AIA9	0	0	1	1	1	1*	0	1	1	0	0	0	0	0	1*	1*	8
10	AIA10	1	1*	1	1*	1	1*	1	1	1	1	0	0	0	0	1*	1*	11
11	AIA11	0	1	1	1	1*	1	1	1	1	1*	0	0	0	0	1*	1*	11
12	AIA12	1	1	1*	1*	1	1	1*	1	1	1	0	1	1	1*	1*	0	14
13	AIA13	1	1	1	1	1	1	1	1	1	1	1	1	1	1*	1*	1*	16
14	AIA14	0	1	1	1	1	1	1	1	1	1	1	0	0	1	1*	1*	13
15	AIA15	1*	1	1*	1*	1*	1	1	1	1	1	1	1	1	0	1	0	14
16	AIA16	0	1*	1*	0	1*	0	1	1	1*	0	0	0	0	0	1*	1	8
	Dependence Power	4	15	16	15	14	14	14	16	15	13	6	6	6	6	16	13	

TABLE VII
TABLE SUMMARIZES THE DIFFERENT LEVELS OF BARRIERS TO AI ADOPTION IN THE HEALTHCARE SECTOR OF INDIA

	Barriers	**Codes**	Partitions level
3	Limited use case of AI applications in healthcare	AIA3	1
4	Transparency and fairness	AIA4	
8	Lack of trust in AI	AIA8	
15	Performance and accuracy of AI-enabled technology	AIA15	
5	Lack of interoperability is a huge limiting factor for AI	AIA5	2
9	High implementation cost	AIA9	
16	Job security and machine-human interactions	AIA16	
2	Government support, regulations,and funding	AIA2	3
7	Data availability and quality challenges	AIA7	
10	Shortage of AI talent causes more delays in adoption	AIA10	
6	AI's potential cybersecurity vulnerabilities	AIA6	4
11	Resistance to change	AIA11	5
12	Lack of clear strategy	AIA12	
13	Limited knowledge and understanding of AI	AIA13	
1	Unavailability of infrastructure and policy support to AI	AIA1	6
14	Lack of leaders' ownership of and commitment to AI	AIA14	

The hierarchical structure of barriers comprises assigned levels, which start from the top and slowly move toward the bottom. The barriers to the bottom of hierarchical matrix are most critical. There are 16 barriers that we considered for this article, which we divided into six levels by employing level partitioning. These level partitions helped us develop the ISM model. The outcomes from the level partitions of barriers to AI adoption in India's healthcare industries are mentioned in Table VII.

e) Interpretive structural modeling (ISM): Further, a conical matrix is developed after concluding the level partitions. The conical matrix exhibits the grades correctly, from level one to level six. The digraph is rendered from the conical matrix. It is a set of nodes indicating the interlinkages between the barriers using arrows. The barrier rates are shown using FRM-based level partition and barrier relationships. Upon completion of the digraph, convert it to an ISM model. The summary of the barrier replaces the digraph nodes. The level one or top-tier barriers are less critical; they become more critical as the level goes higher. The ISM model for AI adoption in India's healthcare industries is shown in Fig. 2 [35], [47], [48].

2) MICMAC Method Analysis: To a great extent, the ISM model was quite informative; we understood the affinities between the stated barriers, whereas MICMAC facilitates the necessary understanding of the level to which one barrier is prejudiced by the other. The analysis was done by contemplating

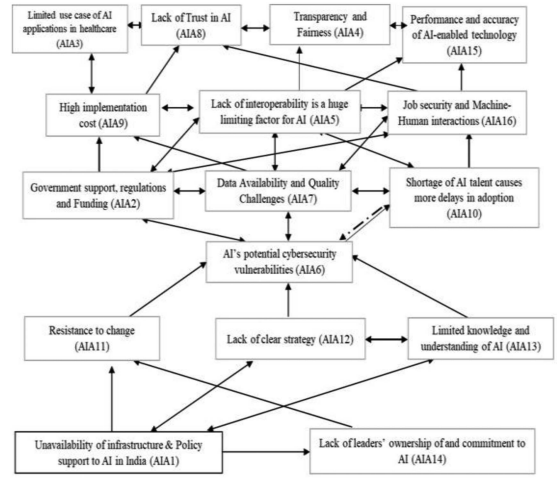

Fig. 2. Figure represents the outcome of the interpretive structural modeling (ISM) for barriers to AI adoption.

the driving and dependence power. The effect of one barrier on another is illustrated through driving power.

Further, a 2-D graph is plotted based on their driving and dependency powers. Barriers are classified into four quadrants based on driving power (Y-axis) and dependence power (X-axis): autonomous barriers (I), dependent barriers (II), linkage barriers (III), and independent barriers (IV) [35], [47], [48].

The barriers in Quadrant I are autonomous barriers. These barriers have weak driving and dependence power. These barriers are disconnected and remain extraneous. Barriers in Quadrant II are dependent barriers with weak driving but strong dependence power. These are influenced by other barriers. The barriers in Quadrant III are linkage barriers with high driving and dependence power. These barriers are highly unstable. Last, the barriers in Quadrant IV are independent barriers with strong driving but weak dependence power. It influences the barriers depended on it and are critical barriers to the process and system. The MICMAC analysis of the barriers that help in the adoption of AI in India's healthcare sector is given in Fig. 3.

IV. RESULTS

Healthcare AI has the potential to transform the healthcare business, saving cost and lives. Despite the public's willingness to participate in AI-assisted healthcare and the elevated investment levels, AI adoption in the healthcare business remains low. It is advised to focus on technological limits, regulatory issues, skepticism, and skills to assist overcome the lack of AI acceptance in healthcare, but little study has addressed the true causes of resistance. AI has shown its importance in every field and is a life-changing technology. By adopting AI early, the industry can leverage its advantage over competitors. In a developing country where resources are scarce, particularly in the healthcare sector, where we have limited medical institutes and doctors, providing healthcare facilities to 1.3 billion people is a big challenge. Technology can play a vital role in catering to this shortage of medical facilities. Still, at the same time, it

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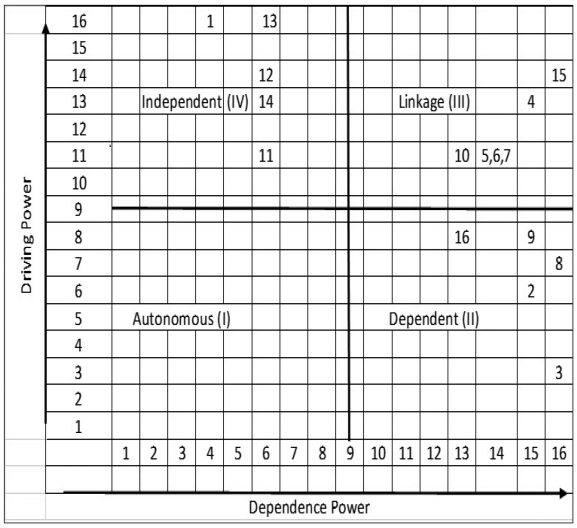

Fig. 3. Figure represents the diagram between dependence power vs. driving power using MICMAC analysis.

cannot allow doctors to diagnose diseases early and precisely and save lives. India is a big market for AI products. The early adoption of AI will benefit physicians and patients and generate new employment and skills, which will further help in nation-building. Though the government of India has already taken many initiatives for AI adoption, many things still need to be done to provide better policy and infrastructure for AI adopters, particularly in the healthcare sector. The AI will also help the government address the issue of providing primary healthcare services to the population and develop healthcare policies based on the insights it gets from technology. We have used the ISM and MICMAC techniques to explore the AI adoption barriers and the relationships between them. The research outcomes from the article are as follows.

1) *Autonomous Barriers:* No barrier falls in Quadrant I, which means no barrier has weak dependence and driving power, and each barrier is important.
2) *Dependent Barriers:* The barriers fall in Quadrant II, a dependent quadrant (2, 3, 8, 9, and 16). This barrier has strong dependence but low driving power. Barrier 2 is government support, regulations, and funding. Barrier 3: limited use case of AI applications in healthcare; Barrier 8: lack of trust in AI; Barrier 9: high implementation cost; and Barrier 16: job security and machine–human interactions are important barriers in the Indian context. Most of the hospitals in India still do not have hospital management systems, and many doctors do not even use computers for prescriptions; they give handwritten prescriptions. In such a scenario, adopting AI is a big challenge. The barrier in Quadrant II shows that, though it has less driving power, AI adoption is not possible without a proper digital platform.
3) *Linkage Barriers:* The barriers (45, 6, 7, 10, and 15) in Quadrant III, which have both strong dependence powers and strong driving powers, are considered the major barriers to AI adoption in healthcare. These barriers eventually influence the other barriers. If one of these problems is solved, the other barriers will also provide some assistance. Following MICMAC analysis, six barriers lie in Quadrant III: barrier 4 (transparency and fairness), barrier 5 (lack of interoperability is a huge limiting factor for AI), barrier 6 (AI's potential cybersecurity vulnerabilities), barrier 7 (data availability and quality challenges), barrier 10 (shortage of AI talent causes more delays in adoption), and barrier 15 (performance and accuracy of AI-enabled technology).
4) *Independent Barriers:* The barriers (1, 11, 12, 13, 14, and 15) in Quadrant IV have strong driving but weak dependence power. This quadrant highlights the key barriers to AI adoption in India's healthcare system. In this article, five barriers lie in this category: barrier 1 (unavailability of infrastructure and policy support for AI in India), barrier 11 (resistance to change), barrier 12 (lack of clear strategy), barrier 13 (limited knowledge and understanding of AI), and barrier 14 (lack of leaders' ownership of and commitment to AI). The above barriers at application/operational/micro label include the following aspects.
   a) *Unavailability of infrastructure and policy support for AI:* These pertain to research capacity infrastructure, regional disparity, lab-based AI intervention, intellectual property constraints, uncertain regulatory, data governance, unregulated training set algorithm, regulatory issues, regulatory channels, copyrights issues, mass-surveillance, bioethical principles, internet connectivity, access to technology, lack of resources, regulatory barriers on safety, barriers to commercialization, hospital resource, education, and treatment [28], [34], [42], [69], [85], [87], [90], [91], [102], [104].
   b) *Lack of leaders' ownership of and commitment to AI:* This includes educate augmented doctors, future medical leaders ambient clinical intelligence, responsibility, and standards, overpromise of service robot capabilities, interdisciplinary research (robotics, computer sciences, and engineering), standards and methods, medical device licensing, rigorous process, time and resource consuming, lack of technological leadership, and lack of accountability [11], [14], [23], [97].
   c) *Limited knowledge and understanding of AI:* Including lack of understanding, communication about technology, problem structuring, and cognitive mapping [39], [68].
   d) *Resistance to change:* Includes psychological resistance, resistance from the workforce, acceptance problem, large variance in acceptance, slow adopters of new technology, relative preference, and propensity to follow [39], [64], [85], [96], [104].
   e) *Lack of clear strategy:* includes misalignment of AI strategies with business strategies, prioritized cost reduction over revenue growth, the divergence of AI initiatives, business transformation efforts, disconnects between reality and expectations, applying AI outside of populations, unrealistic expectations, inadequate evaluation before scaling, lack of top management

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support, organizational and managerial challenges, decision making may be opaque [26], [28], [59], [69], [85], [104].

V. Discussion and Recommendations

The results derived from our exploratory and confirmatory approaches contribute significantly to the practitioners' enactment of IT science, enclosing AIT adoption (or AI technologies) in healthcare and extending the theoretical knowledge surrounding the ISM model. The resulting conclusion leads to distinct commercial challenges: a lack of knowledge of medical professionals' reluctance to use AI may restrict the advantages of investments in AI healthcare technologies. The article recommended the following to the organization based on ISM (Fig. 2) and MICMAC (Fig. 3) outcomes.

A policy-changing mindset to showcase the benefits of AI in terms of time and quality: AI is a rapidly evolving field, and policies need to keep pace with technological advancements. By adopting a policy-changing mindset that recognizes AI's evolving nature and potential benefits in terms of time and quality, policymakers can create an enabling environment for AI innovation while ensuring responsible and ethical deployment. This approach can help showcase the positive impact of AI and maximize its potential for societal and economic growth. A few factors contributing to achieving this goal are continuous policy evaluation and updates; collaboration with stakeholders; agile and adaptive policy frameworks; sandboxing and pilot programs; regulatory frameworks for emerging AI applications; and public awareness and education.

Infrastructure support and definitive policy to boost the commitment of leaders: To promote commitment from leaders and provide infrastructure support for AI adoption, a combination of specific policies and initiatives can be implemented. A definitive policy framework will help guide the adoption of AI, ensuring that its potential benefits in terms of efficiency, productivity, and quality are fully realized. A few initiatives in this direction are establishing AI research and development centers: Investing in AI infrastructure, supporting data sharing and accessibility, fostering AI education and skill development, creating AI innovation zones, developing AI-specific policies and regulations, supporting government-led AI initiatives, and encouraging cross-sector collaboration.

Desire to innovate and expand knowledge base: To foster a desire to innovate and expand the knowledge base in AI adoption, individuals and organizations can take several steps: staying updated with AI advancements; joining AI communities and networks; pursuing AI education and training; fostering a culture of experimentation; collaborating with academia and research institutions; promoting interdisciplinary collaboration; embracing a growth mindset; and engaging in AI competitions and challenges.

Establishing a BCP strategy in a collaborative manner: By establishing a BCP strategy in a collaborative manner, organizations can effectively address the risks and disruptions associated with AI adoption. This collaborative approach ensures that all relevant stakeholders contribute their expertise and insights, leading to a comprehensive and resilient BCP plan. It needs steps like identifying key stakeholders, conducting a risk assessment, defining critical AI systems and processes, and establishing recovery objectives. Developing, testing, validating, documenting, and updating a BCP framework and fostering cross-functional collaboration.

Beta testing to simulate what-if scenario: Beta testing is a valuable approach to simulating what-if scenarios in AI adoption. It allows organizations to gather feedback, identify potential issues, and assess the performance of AI systems in real-world conditions before full deployment. This AI system's performance should be tested under varying workload conditions, unexpected data inputs, or changes in the operational environment. Selecting a diverse user group, encouraging feedback and engagement, prioritizing necessary improvements or adjustments, and iterating on the AI system based on the insights gained from the beta testing phase to enhance its performance, reliability, and adaptability. Beta testing will provide a controlled environment to simulate what-if scenarios, uncover potential issues, and refine the AI system before full deployment. It will allow organizations to gather valuable insights, enhance the user experience, and ensure the AI system's readiness to handle unexpected scenarios in real-world settings.

To be able to see a larger and more comprehensive view: To gain a larger and more comprehensive view of AI adoption, organizations should understand the AI landscape, engage with experts and thought leaders, collaborate with diverse stakeholders, explore use cases across industries, and assess ethical and societal implications.

Toward failsafe system leading to improved data integrity and consistency: Designing failsafe systems for improved data integrity and consistency in AI adoption is crucial for ensuring reliable and trustworthy AI outcomes. This, in turn, will foster greater confidence in AI adoption and will lead to more ethical and responsible use of AI technologies. This may include data auditing, cross-validation, and outlier detection methods.

Agile policymaking to bring expertise together: By adopting agile principles in policymaking, policymakers can effectively bring expertise together, respond to emerging challenges, and create policies that support responsible and beneficial AI adoption. This approach enables flexibility, adaptability, and collaboration, ensuring that policies keep pace with the evolving AI landscape and address the needs of various stakeholders. One of the key steps is adapting the concept of minimum viable policies inspired by minimum viable products in agile software development. The focus must be on enveloping policies that address the most critical and immediate challenges in AI adoption. Starting with policies with the highest impact and feasibility, iterate based on feedback and emerging needs. This allows policymakers to adapt to changing circumstances and make timely adjustments.

Integrated approach toward decision-making: An integrated approach to decision-making in AI adoption will consider multiple dimensions and stakeholders to make informed and responsible decisions. Data-driven insights, reflections on AI adoption's fairness, transparency, accountability, and privacy implications, and risk assessment and mitigation will lead to robust AI adoption.

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Resource sharing and knowledge consolidation: Resource sharing and knowledge consolidation are essential aspects of AI adoption. Open data initiatives, research partnerships, best practices repositories, standards development and adoption, and intellectual property considerations are a few major considerations in this direction.

Concurrent engineering approach to move toward win–win: The concurrent engineering approach can be applied to AI adoption to facilitate a win–win scenario for all stakeholders involved. Concurrent engineering emphasizes the simultaneous and collaborative consideration of various aspects of a project throughout its lifecycle. This approach promotes effective decision-making, risk mitigation, and iterative development, ultimately leading to a win–win outcome, where AI solutions are aligned with business objectives, ethical considerations, and user needs.

Information visibility in interactions: Information visibility is crucial in AI adoption interactions to bring transparency, trust, and effective decision-making. Transparent communication, knowledge sharing, and accountability are essential for responsible and successful AI adoption. Offering transparency mechanisms like data access requests or privacy dashboards will empower individuals to understand and control their data.

Blended approach in decisions: A human–machine blended approach to AI adoption will combine humans' and machines' strengths and capabilities to make better decisions while incorporating human judgment, expertise, and values into decision-making processes. While humans can contribute their critical thinking, intuition, judgment, expertise, values, and creativity to evaluate complex situations and make nuanced judgments, machines, on the other hand, can provide data-driven insights, predictive analytics, and evidence-based recommendations to inform decision-making processes.

Learning curve will reduce mistakes and errors: As organizations and individuals gain experience and knowledge in implementing AI systems, they become more proficient in navigating the challenges and complexities associated with AI adoption. The learning curve in AI adoption is an iterative process where mistakes and errors serve as opportunities for improvement.

Talent mapping, focused training, and succession planning: Talent mapping will entail assessing the existing workforce to determine the availability of AI-related skills and identifying skill gaps. Focused training and hands-on experience will help individuals apply their learning in real-world scenarios and develop proficiency in AI adoption. Providing leadership development programs or mentorship opportunities to groom individuals for future AI leadership roles should focus on developing their strategic thinking, decision-making, project management, and communication skills. Creating career pathways will provide growth opportunities for employees interested in pursuing AI-related roles.

A. Implication and Recommendations for Further Research

The research's theoretical and practical ramifications are noteworthy, offering a methodical framework for comprehending the intricate aspects of AI integration in healthcare and promoting the uptake of AI-based solutions in this field. Research has the potential to improve patient outcomes, lower healthcare costs, and raise the standard of care while encouraging the ethical and responsible use of AI in the medical field.

The research contributes to existing knowledge by identifying the barriers to AI adoption in healthcare, prioritizing these barriers, and developing a conceptual framework for the future implications of AI in the healthcare sector. The research has potential real-world applications in the healthcare context in India, enabling healthcare professionals to leverage AI technologies to raise the standard of treatment, lower healthcare expenses, and improve patient outcomes. The research also has policy implications for the Indian government and other stakeholders, providing a roadmap for AI's responsible and ethical deployment in healthcare.

This article is limited to a few experts and hospitals, which do not represent the population. Researchers may wish to increase the sample size to include all the demographics of India and examine the findings further to understand better the reasons behind some of the unique relationships among the AI adoption barriers. Also, the ISM methods only help show the relationships between the barriers; they do not give any weights or statistically validate the chosen barriers. Therefore, it is suggested that further study be required to test and validate the model by applying the SEM method.

VI. Conclusion

We plan to investigate hospitals' inability to adopt AI, as well as the reasons behind their reluctance. Hospitals' failure to adopt AI prohibits them from taking advantage of potentially life-saving technical advances and cost reductions. Due to the lack of adoption, AI enterprises and IT developers are more inclined to invest in industries with lower entry barriers. This will result in less investments being made for healthcare AI research and fewer opportunities for benefit enhancements. The research outcomes revealed that the unavailability of infrastructure and policy support for AI in India is the major barrier to adopting AI in the healthcare sector. The government must work fast on policy and infrastructure development to accelerate AI adoption.

Data Availability Statement

The data supporting this article's findings are available from the authors upon reasonable request.

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